Digital Image: It is a representation of a scene, in a format that computer can process, consists of grid of tiny units called pixels.

Image Enhancement: Increasing contrast, reducing noise.

Image Restoration: Restore image to its original or an improved state.

Image Compression: Reduce the size of an image file for efficient storage or transmission without losing significant quality.

Image Analysis & Interpretation: Extract meaningful information from an image for further decision-making.

Image segmentation: Divide an image into meaningful regions or segments. for isolation.

Type of processing can be done on digital image?

* Low level processing
  - I/p image.
  - O/p image.
  ex: noise removal.

* Mid level processing:
  * I/p image
  * o/p attribute (edges, contours)
  ex: object segmentation

* High level processing
  - I/p Attributes
  - O/p understanding.
  ex: Scene understanding.

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* **Image Acquisition:** it captures the visual information from the real world & converts it into a digital image.

* **Image:** A 2D projection of 3D object.
  ↓
  $F(x,y)$

Types: **Analog image:** continuous variation of tones.

Digital image: Finite set of pixels.

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Image Enhancement: Improving the visual quality of an image

*Image restoration: process of recovering an image that has been degraded due to various factors such as noise, Loss of information.

*Color image processing: A manipulation & analysis of color image which contain complex information than grayscale image.

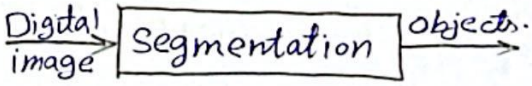
*wavelets & multiresolution processing:
it is transformation of an image into a set of wavelet coefficients, which capture both frequency & spatial information.

*Image compression: process of reducing the size of a digital image.

Types: *Lossless (Exact) ⇒ Original = Decompressed.
*Lossy → Original ≅ Decompressed.

*Morphological processing: A tool for shaped-based image analysis used to extract the boundaries of object in an image

*Image segmentation: Dividing an image into distinct regions or segments to simplify its representation.



→ Most common forms of image representation:

*pixel-Based representation

*vector-Based representation.

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* Relationships: between pixels:

* helps define the structure, content, and image visual quality

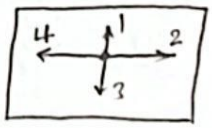
Types: * Neighboring.
* Connectivity.
* Labelling (connected regions)
* Distance measures.

* Neighboring:

- N_4 neighbors (above, below, Left & right)

$$N_4(f(x,y)) = [f(x-1,y), f(x+1,y), f(x,y-1), f(x,y+1)]$$









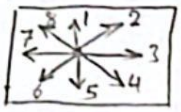
- N_D neighbors (4, Diagonals)

$$N_D(f(x,y)) = [f(x-1,y-1), f(x+1,y-1), f(x-1,y+1), f(x+1,y+1)]$$

- N_8 neighbours (above, left, right, below, & 4 diagonals)

$$N_8(f(x,y)) = [N_4 \cup N_D]$$





⇒ Labelling Alogrithm:

I/P: Labelling particular pixel (row, Column) assume the start from the top left pixel

start: visit pixels from left to right then top to bottom.

if: $f(r,c) \notin$ replace it with 0 and move to next one $f(r,c+1)$.

else: check on the labels of previous neighbor.

if: previous neighbor pixels has the same label then assign the current pixel to same label.

else if: some neighbors in different labels then put pixel in any label of them. and record that all these labels are equivalent

else: start a new lablel & assign that pixel to the new label.

end if: continue to visit until the last pixel visited join all marked equivalent labels to single label

End.

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Distance measures:

→ The most common distance metrices:

- Euclidean distance: A straight-line distance between two pixels.

$$D_e(p,q) = \sqrt{(x-s)^2 + (y-t)^2}$$

- City Block distance (D_4): measure the distance between two pixels by only allowing movement horizontal & vertical directions.

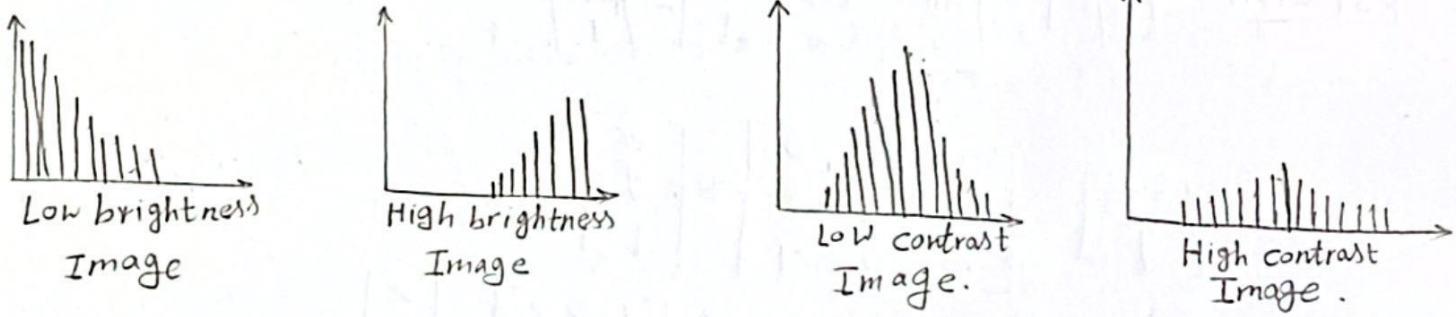
$$D_4(p,q) = |x-s| + |y-t|$$

- Chessboard (D_8): measures distance between two pixels by considering both horizontal & diagonal movements.

$$D_8(p,q) = \max(|x-s|, |y-t|)$$

- Distance measures (D_m): It is defined as the shortest m-path between the pixels.

* **Histogram**: It is a graph, which gives overall idea about the intensity distribution of an image.



* **statistical measures**: provides a quantitative methods for analyzing and interpreting the pixel values within an image.

- gray level values: $L = 0 \rightarrow 255$

max gray level → 50 → Dark Image

minimum gray level 200 → Bright Image

ratio max:min = (100, 150) → low dynamic range.

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Average → 50 → dark Image
→ 200 → bright Image.

variance → high → good contrast
→ Low → low contrast.

* Histogram Equlization: method which increase the dynamic range of the gray levels in low contrast image to cover the Full range of gray levels to improve image contrast.

$$\text{Probability} = P_r(r_k) = \frac{\text{Count}}{M*N}, \quad \text{CDF} = \sum_{j=0}^{k} P_r(r_j)$$

$$\text{Equalized grays} = (L-1) * \sum_{j=0}^{k} P_r(r_j)$$

* Affine Transformation techniques:

Translation: $\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix} * \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$

Rotation: $\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} * \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$

scaling: $\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} S_x & 0 & 0 \\ 0 & S_y & 0 \\ 0 & 0 & 1 \end{bmatrix} * \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$

Flipping. vertical flip: $\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix} * \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$

Horizontal Flip: $\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} * \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$

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Contrast stretching: Low contrast image can result from:

* Poor illumination.
* Lack of dynamic range.

Range points $(r_1, s_1), (r_2, s_2)$ conditions:

$$r_1 > s_1 , r_2 < s_2$$

$$\alpha = \frac{s_1 - 0}{r_1 - 0} , \beta = \frac{s_2 - s_1}{r_2 - r_1} , \gamma = \frac{L-1 - s_2}{L-1 - r_2}$$

$$s = \begin{cases} \alpha * r & 0 \leq r < r_1 \\ \beta(r - r_1) + s_1 & r_1 \leq r < r_2 \\ \gamma(r - r_2) + s_2 & r_2 \leq r < L-1 \end{cases}$$

Image enhancement point processing:

$$S = T(r)$$

S: pixel value of output image after processing.
T: transformation that maps image pixel r into.
r: pixel value of input image before processing.

* **Image Negative:** $S = (L-1) - r$.

* **Logarithmic transformation:** $S = C * \log(1+r)$ where $C = \frac{(L-1)}{\log(L)}$

* **Gamma Correction (power Law):** depend on the value of γ
  * $\gamma < 1$ map a narrow range of dark I/P value
  * $\gamma > 1$ map a narrow range of bright I/P value

$$S = C * r^{\gamma} , C = \frac{255}{255^{\gamma}}$$

* **Gray level ~~scaling~~ slicing:**

[1] Transformation highlights range [A,B] of gray level & reduce all others to constant level. $S(x,y) \begin{cases} 255 & \text{if } A \leq r(x,y) \leq B \\ C & \text{otherwise.} \end{cases}$

[2] Transformation highlights range [A, B] but preserves all other levels.

$S(x,y) \begin{cases} 255 & \text{if } A \leq r(x,y) \leq B. \\ r(x,y) & \text{otherwise} \end{cases}$

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* Digital Image storage & transmission:

* pixel depth = $M * N * K$

* no. of bits required to store the is = $M * N * \log_2(L)$

* Digital image representation:

- **Sampling:** divide the image into samples M & Y & N & X resulting H*N components known as pixels, where each pixel has an intensity (amplitude) value. The result is a grid overbuying the image

- **Quantization:** groups of intervals of intensity into a single value to be a finite number. Digitizing the amplitude value.

* **Dynamic range:** it is the ratio between maximum & minimun intensities in the image (darkest to brightest)

* **Spatial domain:** works directly with pixel values, offering straightforward methods to implement best for local adjustments

* **Frequency domain:** Formed by using Fourier transform, which separates image into their frequency components.

* Types of image processing in spatial domain:

- **Point processing:** focus on individual pixel manipulation through a tranformation function that defines how each pixel value will be modified

- **Neighborhood processing:** focus on modifying pixels value based on its neighbors. The pixel new value is calculated based on the values of itself and its neighbors by using a kernel (mask).

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mparison:

* Histogram equalization:
  - non-linear transformation.
  - Redistributes pixel intensities across the entire range of available values based on cumulative distribution function (CDF)
  - leads to a uniform histogram & enhanced feature visibility.
  - once original pixels are modified, mapping used can't be reversed as it is non-linear.

* Contrast stretching:
  - linear transformation.
  - Linearly maps pixels from a specified range to a wider range.
  - Adjust contrast while preserving overall distribution of original pixel values.
  - Enhances specific regions without affecting others.

- Histogram Equalization is better for low contrast image, while contrast stretching is better for known intensity ranges where a natural look is important.

* Image negative: useful for enhancing white or grey detail embedded in dark regions of an image.

* Logarithmic transformation: Enhancing image details in darker regions while compressing lighter areas.

* Gray level slicing: highlights gray range of interest to the viewer by one of two way (reduce or preserve other levels).

* Bitplane slicing: - represents an image with one or more bits for the byte used for each pixel.
  - converts gray image to binary, image compression (Fewer bits represented)

* Image processing: Image processing modifies the pixels, whereas Computer vision interperts the meaning of those pixels.

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Histogram Algorithm:

Input: Image matrix

start: initialize a frequency array of length based on the pixel depth with Zero values.

$$Frequency[2^{pixel\ depth}] = [0]$$

for all pixels in the image do

Frequency [pixel intensity value] += 1;

end for

Plot the histogram:

use Frequency counts & intensities range to plot the histogram.

Each intensity value is represented as bar, the height of which

End.

Histogram Algorithm:

Input: Image matrix

start: Initialize a Frequency array of length based on the pixel depth with Zero values Frequency $[2^{pixel\ depth}] = \{0\}$

For each pixel in the image matrix do

Frequency [pixel intensity value] += 1;

End for

prob $[2^{pixel\ depth}] = \{0\}$

For i = 0 to $2^{pixel\ depth} - 1$ do

prob[i] = Frequency [i] / M*N

End for

CDF $[2^{pixel\ depth}] = [0]$

CDF[0] = Prob[0]

For i = 1 to $2^{pixel\ depth} - 1$ do

CDF[i] = CDF[i-1] + prob[i]

End for

// Remap normalized intensity to image matri

For each pixel in image matrix do

new - pixel intensity value = round ($2^{pixel\ dept} - 1$ * CDF[pixel intensity Valu

End.

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Image negative Algorithm:

Input: Image matrix

start:

Initialize an output image of the same size as the i/p image

For each pixel in the input image matrix do

Calculate the negative pixel value: $s(x,y) = 255 - r(x,y)$

End.

Gamma Correction Algorithm:

Input: image matrix

start:

Initialize an output image of the same size as the input image

For each pixel in the input image matrix

do

Apply the Gamma Correction Transformation: $s(x,y) = r(x,y)^{1/\gamma}$

normalize the output.

End.

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/ / التاريخ: Image processing. الموضوع:

*Computer Graphics. are concerned with with the creation of Image

*Image processing analyze pictures to derive descriptions (often in mathematical or geometrical forms) of object in pictures.

*Building systems that can automatically analyze, interpret, enhance and manipulate image is the ultimate goal of image processing.

*Image is a 2D projection of a 3D object.

*Image can be defined as a 2D function F(x,y) where x,y represent Spatial coordinates and amplitude F is the intensity.

*Types of images are analog and digital.

*pixel is short for "picture elements".

*Digital image color models are "RGB", "Grayscale", "B&W".

*The first step in image processing that captures info. from the real world and converts it into digital image is acqusition.

*Analog image has variation continuously in tones.

*Digital image is a finite set of digital values called pixels.

*To convert image from analog to digital, we use sampling. and quantization.

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* Sampling is digitizing coordinate value.

* Qunatization is digitizing amplitude value.

* Amplitue values are defined by pixel depth (bits).

* Image enhancement improves visual quality of an image.

* Recovering a degraded image affected by noise, blur or less of information is image restoration.

* color image processing consists of multiple color channel, involves manipulation & analysis of color image.

* wavelets & ~~manipulation~~ multiresolution processing involves transforming an image into a set of wavelet coefficients.

* wavelets coefficients capture frequency & spatial information at multiple scales.

* process of reducing the size of a digital image is image compression.

* Types of compression are lossless & lossy.

* Morphological processing is a tool for shape based image analysis, used to extract boundaries of objects in an image.

* Segmentation involves dividing an image into distinct regions or segment to simplify its representation and make it easier to analyze.

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التاريخ: / / الموضوع: _____

* Feature extraction consists of Feature detection & Feature description

* Feature detection refers to finding features in an image, region or boundary

* Feature description assigns quantitative attributes to the detected features.

* Object ~~description~~ detection involves identifying and classifying objects within an image or video stream.

* The level of detail in an image is called resolution, it's determined by Spatial resolution and intensity resolution.

* Spatial resolution is a measure of the smallest discernible detail in an image. It's expressed as. dpi

* Intensity resolution refers to the smallest discernible charge in intesity level, expressed as bits/pixel (2^k).

* Higher spatial resolution means more detail, sharpness and clarity of image.

* Higher intensity resolution means smoother gradients, better color accuracy and more detailed tones.

* For high resolution image, we must consider memory requirements, prorcessing time & transmission time.

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التاريخ: / / الموضوع: _____

* Dynamic Range is the ratio between maximum and minimum intensities in the image (darkest to brightness)

* pixel relationship defines structure, content, and image visual quality

* Types of relationships between pixels are neighboring, connectivity, labelling, distance measures.

* Two pixels are said to be connected if they're neighbors and belong to the some connectivity set.

* Mixed adjacency is a modification of 8 adjacency and is introduced to eliminate ambiguities in 8-adjacency.

* Distance measures quantify how close or for 2 pixels are.

* most common distance metrics are euclidean, city block D_4, chessboard D_8, D_n.

* Euclidean distance is the straight line distance between 2 pixels (actual distance)

* City block distance measures distance between 2 pixels by allowing horizontal & vertical direction only (maze solving)

* Chessboard considers horizontal/vertical and diagonal movement (pathfinding, pattern recognition, object matching).

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التاريخ: / / الموضوع: _____

* D_m distance is the ~~fastest~~ shortest m-path between the pixels, depends on pixel values and their neighbors values.

* Statistical measures provide a quantitative method for analyzing pixel values within an image.

* IF ratio between max & min = (100, 150) → Low dynamic range.

* Image enhancement is done by combining geometric transformation and enhancement techniques using cascading operations.

* Geometric Transformation changes spatial arrangement of pixels to correct distortions that occur with non ideal camera angles.

* Enhancement modifies pixel values without changing their spatial relationships, performed in spatial frequency domain.

* Types of geometric transformations are affine and non-affine.

* Translations shifts image content along x or y axes by specified amounts.

* Rotations rotates image around a specified point by a given angle.

* Scaling changes image size by a scaling factor which can be uniform

* Flipping flips image over specified axis.

* skewing deforms shape like shifted slices.

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*Agraph that gives you an overall idea about intensity distributions of an image is histogram.

*Spatial damain is best for local adjustments, Frequency domain is best for global adjustments.

*Fourier transform separates image into its frequency components

*Image enhancement is categorized into point processing and neighborhood processing

*pointprocessing focuses on individual pixel manipulation.

*Neighborhood focuses on modifying a pixel's value based on its neighbors, use a kernel.

*point processing techniques are histogram equalization, contrast stretching image negative, logarithmic transformation...

*Histogram equalization increases dynamic range of gray levels in a low contrast image to cover full range to improve contrast.

*Histogram equalization improves dark or washed out images.

*Low contrast results from poor illumination and lack of dynamic range.

*Histogram equalization is non-linear, while contrast stretching is linear

التاريخ: / / الموضوع:

*when image is resized to a smaller dimension, pixel quality decreases due to loss of detail.

*A higher dynamic range allows for more detail in shadows and highlight

*Image analysis involves measuring and characterizing properties of an image.

*Lossy compression can significally reduce file size.

*Color is represented in a true color image using 3 color channels (RGB)

*pixel depth is number of bits used to represent color of a single pixe

*neighboring pixels are pixels that share a common boundary or adjacent to eachother.

*we use histogram equalization instead of contrast stretching if the image has limited dynamic range.

*Image negative is useful for enhancing white or grey detail embedded in dark regions of an image.

*Logarithmic transformation enhances image details in a darker regions while ~~processing~~ Compressing ~~higher~~ lighter areas.

*Grey level slicing. hilight gray range of interest to a viewer, Highlight + reduce or highlight + preserve.

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* Bit plane slicing represents an image with one or more bits of byte used for each pixel.

* Main goal of bit plane slicing converting gray levels image to a binary image, image compression (Fewer bits).

* Digital image processing is a set of ~~bright~~ algorithms applied to improve quality, storage, transmission and representation.

* Things that only neighborhood processing can do are smoothing and edge detection. (sharpening).

* Each pixel's new value is computed as a weighted average of its neighboring pixels, defined by the kernel.

* Types of smoothing filters are averaging and weighted averaging.

* Size of filter kernel affects smoothing degree, where larger kernels yields to more smoothing but more edge blurring (Averaging).

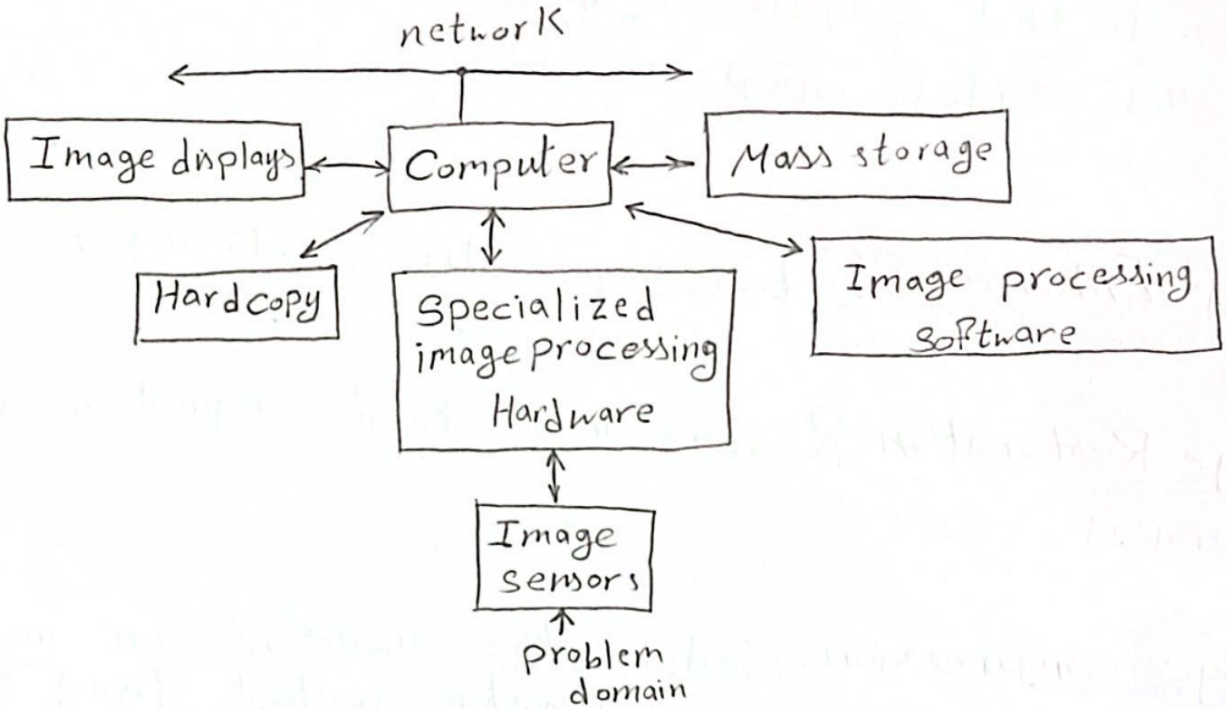
* Averaging filter averages pixel values within a neighborhood.

* weighted averaging filter assigns different weights to pixels in the neighborhood, mean some pixels contribute more than others.

* ~~for~~ weights can be based on distance from center pixel or gaussian distribution.

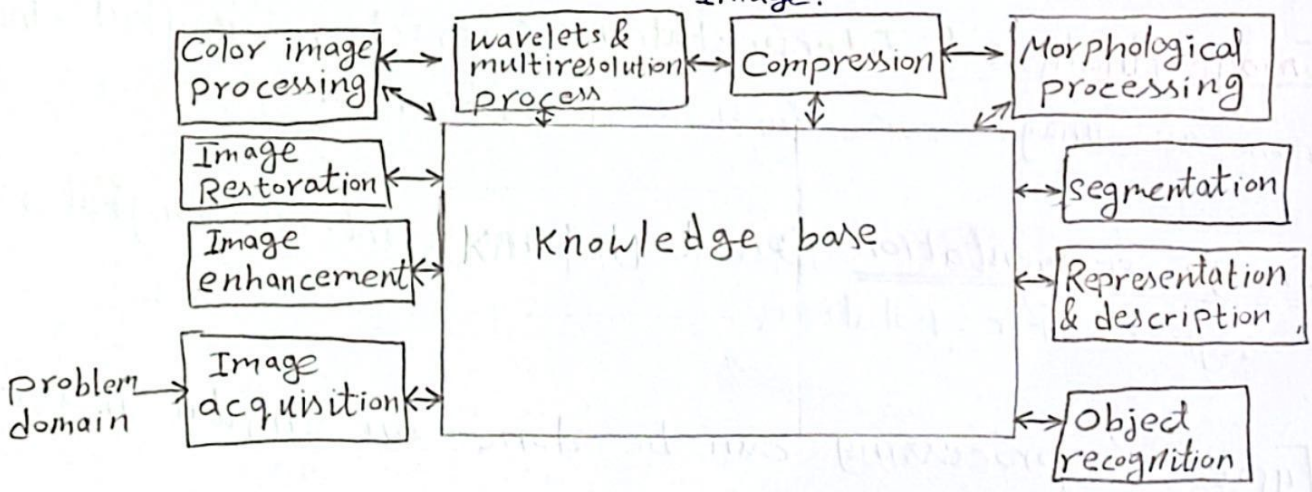
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Basic components of DIP system:



Fundamental steps in DIP:

o/p of these processes ~~are~~ generally are Image.



o/p of these processes ~~are~~ generally are Attributes

*Image Acquisition: it captures the visual information from

* weighted averaging filter allows for edge preservation while still reducing noise.

Computer graphics Vs Image processing.

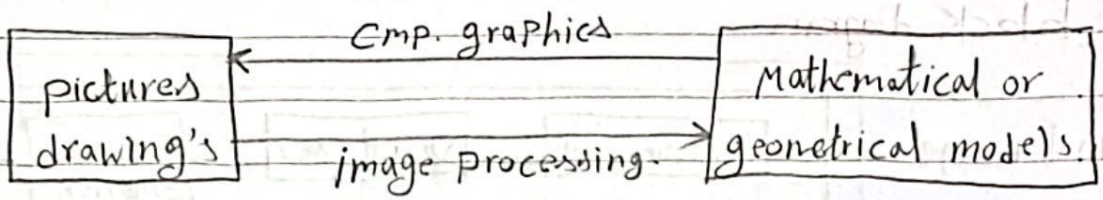
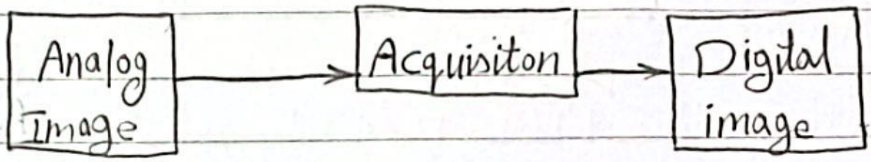
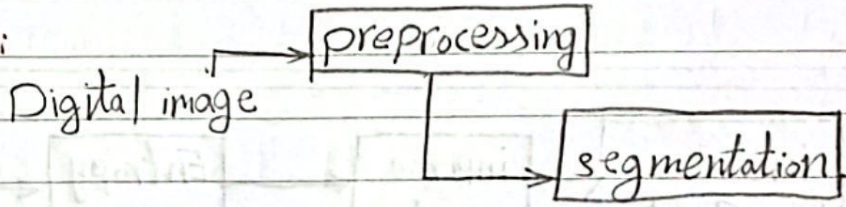


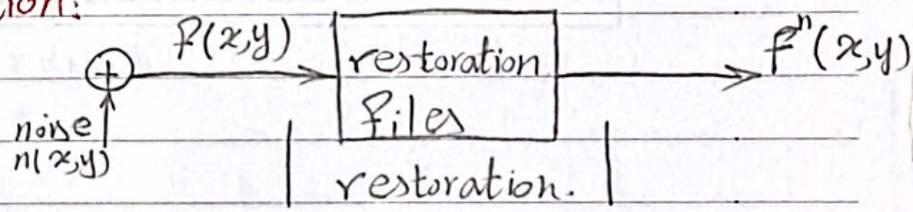
Image acquisition:



segmentation:



Restoration & reconstruction:



Frequency domain filtering operation.

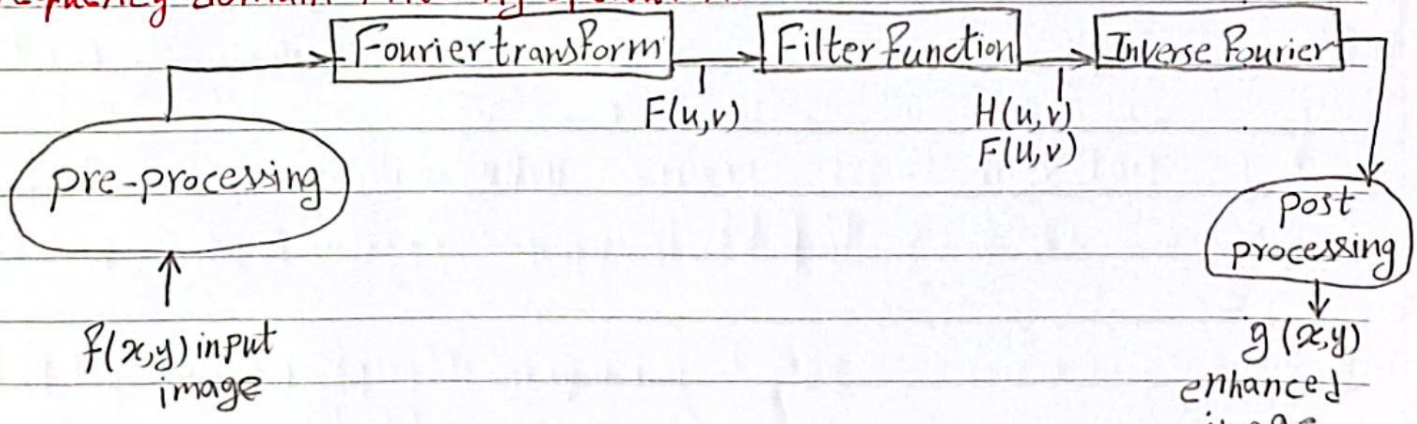
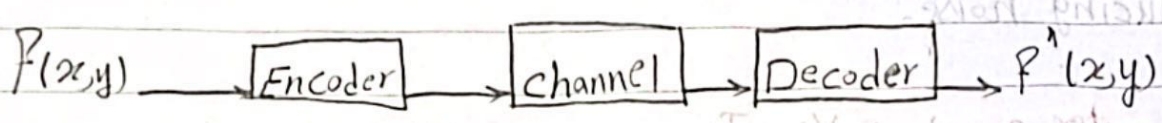


Image compression model:



DCT block diagram.

